



TEST REPORT EN IEC 60086-4: 2019

Primary batteries

Part 4: Safety of lithium batteries

Report Reference No...... : ZKT-2204122506S

Date of issue..... : Apr. 19, 2022

Total number of pages..... : 22

Testing Laboratory..... : Shenzhen ZKT Technology Co., Ltd.

Address..... : 1/F, No. 101, Building B, No. 6, Tangwei Community Industrial Avenue, Fuhai Street, Bao'an District, Shenzhen, China

Applicant's name..... : Luoyang Anxin New Energy Technology Co. LTD

Address..... : Anxin Industrial Park, Dongdajiao Village, Luolong District, Luoyang city, Henan Province

Test specification:

Standards..... : IEC 60086-4: 2019

Test procedure..... : N/A

Non-standard test method..... : N/A

Test Report Form No..... : TRF-EL-061_V0

Test Report Form(s) Originator..... : ZKT Testing

Master TRF..... : Dated: 2020-01-06

This test report is specially limited to the above client company and product model only. It may not be duplicated without prior written consent of ZKT Test.

Test item description..... : Lithium manganese buckle battery CR2032

Trade Mark..... : ETEnergy

Manufacturer..... : Same as applicant

Model/Type reference..... : CR2032

Ratings..... : 3V, 220mAh



Testing procedure and testing location:

Testing Laboratory.....: Shenzhen ZKT Technology Co., Ltd.

**Address.....: 1/F, No. 101, Building B, No. 6, Tangwei Community
Industrial Avenue, Fuhai Street, Bao'an District,
Shenzhen, China**

Date of Test.....: Apr. 12, 2022 to Apr. 19, 2022

Tested by (name + signature).....: Peter Huang

Reviewed by (name + signature).....: Simon Gong

Approved by(name + signature).....: Awen He



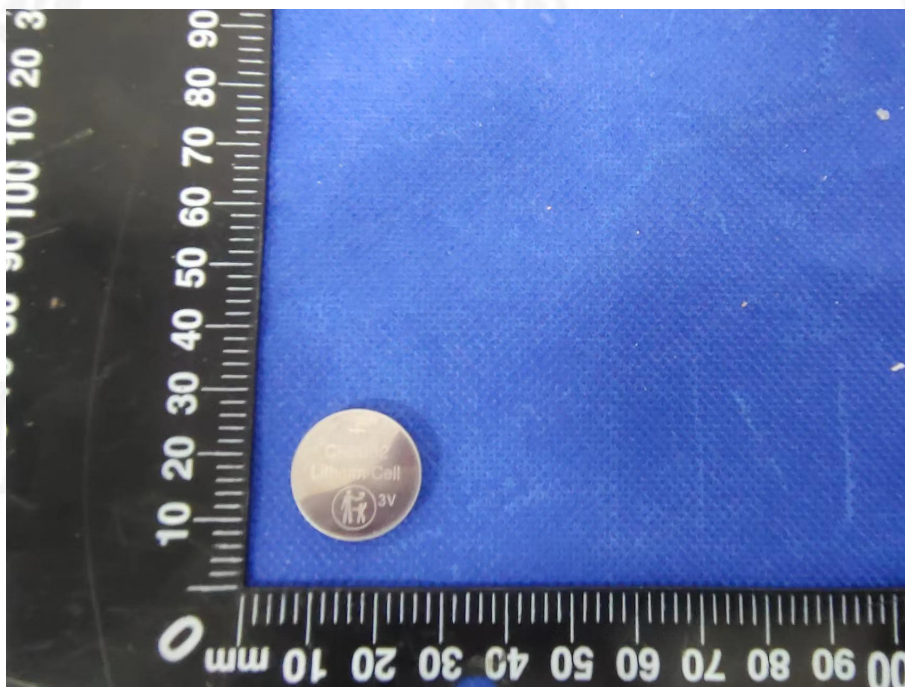


Testing procedure and testing location:	
Testing Laboratory:	
Testing location/ address..... : Shenzhen ZKT Technology Co., Ltd. 1/F, No. 101, Building B, No. 6, Tangwei Community Industrial Avenue, Fuhai Street, Bao'an District, Shenzhen, China	
List of Attachments:	
Appendix 1: 1 page of Photo Documentation	
Summary of testing:	
Tests performed (name of test and test clause): Clause 6.4.1 Test A: Altitude; Clause 6.4.2 Test B: Thermal cycling; Clause 6.4.3 Test C: Vibration; Clause 6.4.4 Test D: Shock; Clause 6.5.1 Test E: External short-circuit; Clause 6.5.3 Test G: Crush; Clause 6.5.4 Test H: Forced discharge; Clause 6.5.5 Test I: Abnormal charging; Clause 6.5.6 Test J: Free fall; Clause 6.5.7 Test K: Thermal abuse. Clause 6.5.8 Test L: Incorrect installation	Testing location: Shenzhen ZKT Technology Co., Ltd. 1/F, No. 101, Building B, No. 6, Tangwei Community Industrial Avenue, Fuhai Street, Bao'an District, Shenzhen, China
Summary of compliance with National Differences	
N/A	
☒ The product fulfils the requirements of EN IEC 60086-4: 2019	



Copy of marking plate

The artwork below may be only a draft. The use of certification marks on a product must be authorized by the respective NCBs that own these marks.





Test item particulars..... :	
Classification of installation and use.....:	To be defined in final product
Supply connection.....:	Custom output
Weight..... :	2.906g
Possible test case verdicts:	
- test case does not apply to the test object..... :	N/A
- test object does meet the requirement..... :	P (Pass)
- test object does not meet the requirement..... :	F (Fail)
Testing..... :	
Date of receipt of test item..... :	Apr. 12, 2022
Date (s) of performance of tests..... :	Apr. 12, 2022 to Apr. 19, 2022
General remarks:	
The test results presented in this report relate only to the object tested. This report shall not be reproduced, except in full, without the written approval of the Issuing testing laboratory. "(See Enclosure #)" refers to additional information appended to the report. "(See appended table)" refers to a table appended to the report. Throughout this report a ☐ comma / ☒ point is used as the decimal separator.	
Name and address of factory (ies) : Luoyang Anxin New Energy Technology Co. LTD Anxin Industrial Park, Dongdajiao Village, Luolong District, Luoyang city, Henan Province	



General product information:

This sample is constructed with single lithium metal cells

The main features of the cell in the battery are shown as below

Model	Nominal Capacity	Nominal Voltage	Maximum Discharge Current	Final Voltage
CR2032	220mAh	3.0V	5mA	2V



EN IEC 60086-4: 2019			
Clause	Requirement + Test	Result - Remark	Verdict
4	REQUIREMENTS FOR SAFETY		P
4.1	Design		P
	a) Abnormal temperature rise above the critical value prevented		P
	b) Temperature increases in the battery controlled		P
	c) Lithium manganese buckle battery CR2032s and batteries designed to relieve excessive internal pressure or to preclude a violent rupture under conditions of transport, intended use and reasonably foreseeable misuse	Pressure relieving device exists.	P
4.2	Quality plan		P
	Manufacturer prepared and implemented a quality plan defining the procedures for the inspection of materials, components, cells and batteries during the course of manufacture, to be applied to the total process of producing a specific type of battery	Complied. Battery factory provided Quality plan. Cell manufacturer provided ISO 9001:2015 cert.	P
	Manufactures understood their process capabilities and instituted the necessary process controls as they relate to product safety		P
5	TYPE TESTING AND SAMPLING		P
5.1	Validity of testing		P
5.2	Test samples		P
6	TESTING AND REQUIREMENTS		P
6.1	General	Complied.	P
6.1.1	Test application matrix	(See 6.2)	P
	s: cell or single cell battery	Cell	P
	m: multi cell battery		N/A
6.1.3	Ambient temperature (°C)	20°C ± 5°C.	P
6.1.4	Parameter measurement tolerances		P
6.1.5	Predischarge		P
6.1.6	Additional cells		P
6.2	Evaluation of test criteria		P
6.2.1	Short-circuit		P
6.2.2	Excessive temperature rise		P
6.2.3	Leakage		P
6.2.4	Venting		P
6.2.5	Fire		P
6.2.6	Rupture		P



EN IEC 60086-4: 2019			
Clause	Requirement + Test	Result - Remark	Verdict
6.2.7	Explosion		P

6.3	Tests and requirements – Overview	(See table 4 in the standard)	P
6.4	Tests for intended use		P
6.4.1	Test A: Altitude	(See appended Table 1 and Table 6.4.1)	P
6.4.2	Test B: Thermal cycling	(See appended Table 1 and Table 6.4.2)	P
6.4.3	Test C: Vibration	(See appended Table 1 and Table 6.4.3)	P
6.4.4	Test D: Shock	(See appended Table 1 and Table 6.4.4)	P
6.5	Tests for reasonably foreseeable misuse		P
6.5.1	Test E: External short-circuit	(See appended Table 1 and Table 6.5.1)	P
6.5.2	Test F: Impact	(See appended Table 1 and Table 6.5.2)	N/A
6.5.3	Test G: Crush	(See appended Table 1 and Table 6.5.3)	P
6.5.4	Test H: Forced discharge	(See appended Table 1 and Table 6.5.4)	P
6.5.5	Test I: Abnormal charging	(See appended Table 1 and Table 6.5.5)	P
6.5.6	Test J: Free fall	(See appended Table 1 and Table 6.5.6)	P
6.5.7	Test K: Thermal abuse	(See appended Table 1 and Table 6.5.7)	P
6.5.8	Test L: Incorrect installation	(See appended Table 1 and Table 6.5.8)	P
6.5.9	Test M: Overdischarge	(See appended Table 1 and Table 6.5.9)	N/A
6.6	Information given in the relevant specification		P
	a) PredischARGE current or resistive load and end-point voltage specified by the manufacturer.....	Discharging current: 0.5mA; End point voltage: 2.0V	P
	b) Method to measure the energy of an explosion, if any		N/A
	c) Shape: prismatic, flexible, coin or cylindrical Diameter: less than 18 mm or not less than 18 mm	Coin cell, Diameter is 20mm, not less than 18 mm.	P
	d) Maximum continuous discharge current specified by the manufacturer for test H.....	Max. continuous discharge current: 5mA.	P
	e) Rated capacity specified by the manufacturer for test H.....	Rated Capacity: 220mAh.	P



EN IEC 60086-4: 2019			
Clause	Requirement + Test	Result - Remark	Verdict
	f) Abnormal charging current declared by the manufacturer for test I	0.5mA as specified by manufacturer applied.	P
	g) Normal reverse current declared by the manufacturer which applied to the battery during its operating life.....		P
7	INFORMATION FOR SAFETY		P
7.1	Safety precautions during design of equipment		P
7.1.1	General		P
7.1.2	Charge protection	Considered in end product.	N/A
7.1.3	Parallel connection	Considered in end product.	N/A
7.2	Precautions during handling of batteries	Information for safety mentioned in manufacturer's specifications.	P
7.3	Packaging		P
7.4	Handling of battery cartons		N/A
7.5	Transport		P
7.5.1	General		P
7.5.2	Air transport		P
7.5.3	Sea transport		P
7.5.4	Land transport		P
7.6	Display and storage	Information for display and storage mentioned in manufacturer's specifications.	P
7.7	Disposal	Information for disposal instructions mentioned in manufacturer's specifications.	P
8	INSTRUCTIONS FOR USE	Information for use mentioned in manufacturer's specifications.	P
9	MARKING AND PACKAGING		P
9.1	General	See the label.	P
9.2	Swallowable batteries		N/A
9.3	Safety pictograms		P
ANNEX A	(INFORMATIVE) GUIDELINES FOR THE ACHIEVEMENT OF SAFETY OF LITHIUM BATTERIES		N/A
ANNEX B	(INFORMATIVE) GUIDELINES FOR DESIGNERS OF EQUIPMENT USING LITHIUM BATTERIES		N/A
ANNEX C	(INFORMATIVE) ADDITIONAL INFORMATION ON DISPLAY AND STORAGE		N/A



EN IEC 60086-4: 2019			
Clause	Requirement + Test	Result - Remark	Verdict
ANNEX D	(INFORMATIVE) SAFETY PICTOGRAMS		N/A
D.1	General		N/A
D.2	Pictograms		N/A
D.3	Instruction for use		N/A
ANNEX E	(NORMATIVE) CHILD RESISTANT PACKAGING OF COIN CELLS		N/A
E.1	General		N/A
E.2	Applicability		N/A
E.3	Packaging tests		N/A
E.3.1	General		N/A
E.3.2	Test items		N/A
E.3.3	Test procedure		N/A
E.3.4	Criteria		N/A
ANNEX F	(INFORMATIVE) USE OF THE KEEP OUT OF REACH OF CHILDREN SAFETY SIGN		N/A
F.1	General		N/A
F.2	Safety sign		N/A
F.3	Best practices for marking the packaging		N/A
F.4	Best practices for marking the cell		N/A



TABLE: Critical components information					P
Object / part No.	Manufacturer / trademark	Type / model	Technical data	Standard	Mark(s) of conformity ¹⁾
Cell	EVERWIN INDUSTRY CO., LTD	CR2032	3V, 220mAh	EN IEC 60086-4: 2019	Tested with appliance
-Positive electrode	Tianjingzhongneng Ltd	Lithium pieces	Li Specific capacity: 3846 mAh/g	--	--
-Negative electrode	Hunan xingtian Ltd	EMD	MnO ₂ , Dimensions: 11 mm * 1,1 mm Specific capacity: 308 mAh/g	--	--
-Separator	Lixin technology	pp	PP Dimensions: 11 mm * 0,10 mm Shut down temperature: 140° C	--	--
-Electrolyte	Factory preparation	--	LiClO ₄ Conductivity: 9,6 mS/cm	--	--
Supplementary information: ¹⁾ Provided evidence ensures the agreed level of compliance. See OD-CB2039.					



TABLE 1 (clause 6.4.1 – 6.5.9)					
Tests	Cell / battery type	Discharge state	Number of test sample	Test result	Verdict
A to D	Cells and single cell batteries	Undischarged	10	NL, NV, NC, NR, NE, NF	P
		Fully discharged	10	NL, NV, NC, NR, NE, NF	P
	Multi-cell batteries	Undischarged	4		N/A
		Fully discharged	4		N/A
E	Cells and single cell batteries	Undischarged	10	NT, NR, NE, NF	P
		Fully discharged	10	NT, NR, NE, NF	P
	Multi-cell batteries	Undischarged	4		N/A
		Fully discharged	4		N/A
F	Cells and single cell batteries	Undischarged	5	NT, NR, NE, NF	P
		Fully discharged	5	NT, NR, NE, NF	P
	Multi-cell batteries	Undischarged	5 component cells		N/A
		Fully discharged	5 component cells		N/A
G	Cells and single cell batteries	Undischarged	5	NT, NE, NF	P
		Fully discharged	5	NT, NE, NF	P
	Multi-cell batteries	Undischarged	5 component cells		N/A
		Fully discharged	5 component cells		N/A
H	Cells and single cell batteries	Fully discharged	10	NE, NF	P
	Multi-cell batteries	Fully discharged	10 component cells		N/A
I	Cells and single cell batteries	Undischarged	5	NE, NF	P
	Multi-cell batteries	Undischarged	5		N/A
J	Cells and single cell batteries	Undischarged	5	NV, NE, NF	P
	Multi-cell batteries	Undischarged	5		N/A
K	Cells and single cell batteries	Undischarged	5	NE, NF	P
	Multi-cell batteries	Undischarged	5		N/A
L	Cells and single cell batteries	Undischarged	20	NE, NF	P
M	Cells and single cell batteries	50 % predischarged	20		N/A
		75 % predischarged	20		N/A



Supplementary information:

NC: No short-circuit

NE: No explosion

NF: No fire

NL: No leakage

NR: No rupture

NT: No excessive temperature rise

NV: No venting

DRAFT



6.4.1	TABLE: Test A: Altitude (Undischarged)			P
Battery No.	Before test		After test	
	battery voltage (V)	battery weight (g)	battery volt age (V)	battery weight (g)
CR2032-C1#	3.26	2.907	3.26	2.906
CR2032-C2#	3.28	2.902	3.27	2.902
CR2032-C3#	3.25	2.911	3.25	2.911
CR2032-C4#	3.26	2.908	3.25	2.907
CR2032-C5#	3.21	2.903	3.20	2.903
CR2032-C6#	3.28	2.904	3.28	2.904
CR2032-C7#	3.27	2.904	3.27	2.903
CR2032-C8#	3.20	2.908	3.20	2.908
CR2032-C9#	3.26	2.906	3.26	2.906
CR2032-C10#	3.29	2.907	3.28	2.906

Supplementary information:

- No leakage, no venting, no short-circuit, no rupture, no explosion and no fire.

6.4.1	TABLE: Test A: Altitude (Fully discharged)			P
Battery No.	Before test		After test	
	battery voltage (V)	battery weight (g)	battery voltage (V)	battery weight (g)
CR2032-C11#	2.89	2.905	2.88	2.905
CR2032-C12#	2.92	2.903	2.92	2.902
CR2032-C13#	2.86	2.912	2.86	2.912
CR2032-C14#	2.91	2.914	2.91	2.913
CR2032-C15#	2.88	2.905	2.88	2.905
CR2032-C16#	2.90	2.904	2.90	2.903
CR2032-C17#	2.86	2.908	2.86	2.908
CR2032-C18#	2.86	2.906	2.86	2.906
CR2032-C19#	2.92	2.911	2.92	2.911
CR2032-C20#	2.87	2.915	2.87	2.914

Supplementary information:

- No leakage, no venting, no short-circuit, no rupture, no explosion and no fire.

6.4.2	TABLE: Test B: Thermal cycling (Undischarged)			P
Battery No.	Before test		After test	
	battery voltage (V)	battery weight (g)	battery voltage (V)	battery weight (g)
CR2032-C1#	3.26	2.906	3.25	2.905
CR2032-C2#	3.27	2.905	3.26	2.905
CR2032-C3#	3.25	2.914	3.25	2.913



CR2032-C4#	3.25	2.912	3.25	2.912
CR2032-C5#	3.20	2.908	3.20	2.908
CR2032-C6#	3.28	2.908	3.27	2.906
CR2032-C7#	3.27	2.907	3.26	2.907
CR2032-C8#	3.20	2.906	3.20	2.906
CR2032-C9#	3.26	2.913	3.25	2.912
CR2032-C10#	3.28	2.909	3.28	2.909

Supplementary information:

- No leakage, no venting, no short-circuit, no rupture, no explosion and no fire.

6.4.2	TABLE: Test B: Thermal cycling (Fully discharged)			P
Battery No.	Before test		After test	
	battery voltage (V)	battery weight (g)	battery voltage (V)	battery weight (g)
CR2032-C11#	2.88	2.908	2.87	2.907
CR2032-C12#	2.92	2.906	2.92	2.906
CR2032-C13#	2.86	2.916	2.86	2.915
CR2032-C14#	2.91	2.911	2.91	2.911
CR2032-C15#	2.88	2.908	2.88	2.908
CR2032-C16#	2.90	2.908	2.90	2.907
CR2032-C17#	2.86	2.907	2.86	2.907
CR2032-C18#	2.86	2.906	2.86	2.906
CR2032-C19#	2.92	2.912	2.92	2.911
CR2032-C20#	2.87	2.909	2.87	2.908

Supplementary information:

- No leakage, no venting, no short-circuit, no rupture, no explosion and no fire.

6.4.3	TABLE: Test C: Vibration (Undischarged)			P
Battery No.	Before test		After test	
	battery voltage (V)	battery weight (g)	battery voltage (V)	battery weight (g)
CR2032-C1#	3.25	2.912	3.25	2.911
CR2032-C2#	3.26	2.908	3.25	2.907
CR2032-C3#	3.25	2.911	3.25	2.911
CR2032-C4#	3.25	2.910	3.24	2.910
CR2032-C5#	3.20	2.905	3.20	2.904
CR2032-C6#	3.27	2.908	3.26	2.908
CR2032-C7#	3.26	2.903	3.25	2.902
CR2032-C8#	3.20	2.908	3.20	2.908
CR2032-C9#	3.25	2.914	3.25	2.913
CR2032-C10#	3.28	2.906	3.27	2.905

**Supplementary information:**

- No leakage, no venting, no short-circuit, no rupture, no explosion and no fire.

6.4.3	TABLE: Test C: Vibration (Fully discharged)			P
Battery No.	Before test		After test	
	battery voltage (V)	battery weight (g)	battery voltage (V)	battery weight (g)
CR2032-C11#	2.87	2.912	2.87	2.911
CR2032-C12#	2.92	2.908	2.92	2.908
CR2032-C13#	2.86	2.908	2.86	2.907
CR2032-C14#	2.91	2.916	2.91	2.916
CR2032-C15#	2.88	2.904	2.88	2.904
CR2032-C16#	2.90	2.908	2.90	2.908
CR2032-C17#	2.86	2.904	2.86	2.903
CR2032-C18#	2.86	2.905	2.86	2.905
CR2032-C19#	2.92	2.912	2.92	2.911
CR2032-C20#	2.87	2.908	2.87	2.908

Supplementary information:

- No leakage, no venting, no short-circuit, no rupture, no explosion and no fire.

6.4.4	TABLE: Test D: Shock (Undischarged)			P
Battery No.	Before test		After test	
	battery voltage (V)	battery weight (g)	battery voltage (V)	battery weight (g)
CR2032-C1#	3.25	2.905	3.24	2.905
CR2032-C2#	3.25	2.905	3.25	2.904
CR2032-C3#	3.25	2.913	3.25	2.913
CR2032-C4#	3.24	2.912	3.24	2.912
CR2032-C5#	3.20	2.908	3.20	2.908
CR2032-C6#	3.26	2.906	3.26	2.906
CR2032-C7#	3.25	2.907	3.25	2.905
CR2032-C8#	3.20	2.906	3.20	2.906
CR2032-C9#	3.25	2.912	3.25	2.911
CR2032-C10#	3.27	2.909	3.27	2.909

Supplementary information:

- No leakage, no venting, no short-circuit, no rupture, no explosion and no fire.



6.4.4	TABLE: Test D: Shock (Fully discharged)			P
Battery No.	Before test		After test	
	battery voltage (V)	battery weight (g)	battery voltage (V)	battery weight (g)
CR2032-C11#	2.87	2.911	2.87	2.911
CR2032-C12#	2.92	2.908	2.92	2.908
CR2032-C13#	2.86	2.907	2.86	2.907
CR2032-C14#	2.91	2.916	2.91	2.915
CR2032-C15#	2.88	2.904	2.88	2.903
CR2032-C16#	2.90	2.908	2.90	2.908
CR2032-C17#	2.86	2.903	2.86	2.903
CR2032-C18#	2.86	2.905	2.86	2.905
CR2032-C19#	2.92	2.911	2.92	2.911
CR2032-C20#	2.87	2.908	2.87	2.907
Supplementary information: - No leakage, no venting, no short-circuit, no rupture, no explosion and no fire.				

6.5.1	TABLE: Test E: External short-circuit (Undischarged)				P
Battery No.	Ambient, (°C)	OCV at start of test, (Vdc)	Resistance of circuit, (mΩ)	Maximum case temperature (°C)	Results
CR2032-C1#	55.5	3.23	81.2	90.6	P
CR2032-C2#	55.5	3.22	82.0	92.7	P
CR2032-C3#	55.5	3.23	81.5	93.3	P
CR2032-C4#	55.5	3.25	81.6	91.8	P
CR2032-C5#	55.5	3.23	82.2	92.5	P
CR2032-C6#	55.5	3.21	81.7	93.1	P
CR2032-C7#	55.5	3.22	82.5	91.8	P
CR2032-C8#	55.5	3.22	82.6	92.9	P
CR2032-C9#	55.5	3.23	81.5	93.0	P
CR2032-C10#	55.5	3.25	82.3	92.5	P

Supplementary information:

- No excessive temperature rise (>170°C), no rupture, no explosion and no fire.

6.5.1	TABLE: Test E: External short-circuit (Fully discharged)				P
Battery No.	Ambient, (°C)	OCV at start of test, (Vdc)	Resistance of circuit, (mΩ)	Maximum case temperature (°C)	Results
CR2032-C11#	55.6	2.82	82.6	55.8	P
CR2032-C12#	55.6	2.85	81.3	55.8	P
CR2032-C13#	55.6	2.88	82.2	55.7	P



CR2032-C14#	55.6	2.86	82.7	55.6	P
CR2032-C15#	55.6	2.81	81.5	55.6	P
CR2032-C16#	55.6	2.80	82.8	55.8	P
CR2032-C17#	55.6	2.85	81.6	55.6	P
CR2032-C18#	55.6	2.92	82.3	55.8	P
CR2032-C19#	55.6	2.87	82.7	56.0	P
CR2032-C20#	55.6	2.90	83.5	55.8	P
Supplementary information: - No excessive temperature rise (>170°C), no rupture, no explosion and no fire.					

6.5.2	TABLE: Test F: Impact (Undischarged)				N/A
Battery No.	Ambient, (°C)	OCV at start of test, (Vdc)	Maximum case temperature (°C)	Results	
Supplementary information:					
- No excessive temperature rise (>170°C), no explosion and no fire.					

6.5.2	TABLE: Test F: Impact (Fully discharged)				N/A
Battery No.	Ambient, (°C)	OCV at start of test, (Vdc)	Maximum case temperature (°C)	Results	
Supplementary information:					
- No excessive temperature rise (>170°C), no explosion and no fire.					

6.5.3	TABLE: Test G: Crush (Undischarged)				P
Battery No.	Ambient, (°C)	OCV at start of test, (Vdc)	Maximum case temperature (°C)	Results	
CR2032-C21#	23.3	3.25	23.4	P	
CR2032-C22#	23.3	3.22	23.4	P	
CR2032-C23#	23.3	3.23	23.3	P	



CR2032-C34#	23.3	3.24	23.5	P
CR2032-C25#	23.3	3.23	23.4	P

Supplementary information:

- No excessive temperature rise (>170°C), no explosion and no fire.

6.5.3	TABLE: Test G: Crush (Fully discharged)			P
Battery No.	Ambient, (°C)	OCV at start of test, (Vdc)	Maximum case temperature (°C)	Results
CR2032-C26#	23.5	2.91	23.3	P
CR2032-C27#	23.5	2.90	23.3	P
CR2032-C28#	23.5	2.88	23.6	P
CR2032-C29#	23.5	2.89	23.6	P
CR2032-C30#	23.5	2.90	23.6	P

Supplementary information:

- No excessive temperature rise (>170°C), no explosion and no fire.

6.5.4	TABLE: Test H: Forced discharge (Fully discharged)			P
Battery No.	OCV at start of test, (Vdc)	Max. discharge current, (mA)	Test duration, (Hours)	Results
CR2032-C31#	3.23	5	1	P
CR2032-C32#	3.28	5	1	P
CR2032-C33#	3.25	5	1	P
CR2032-C34#	3.20	5	1	P
CR2032-C35#	3.26	5	1	P
CR2032-C36#	3.22	5	1	P
CR2032-C37#	3.25	5	1	P
CR2032-C38#	3.23	5	1	P
CR2032-C39#	3.26	5	1	P
CR2032-C40#	3.27	5	1	P

Supplementary information:

- No explosion and no fire.

6.5.5	TABLE: Test I: Abnormal charging (Undischarged)			P
Battery No.	OCV at start of test, (Vdc)	Max. charge current, (mA)	Test duration, (Hours)	Results
CR2032-C41#	3.23	5	4.17	P
CR2032-C42#	3.25	5	4.17	P
CR2032-C43#	3.26	5	4.17	P
CR2032-C44#	3.25	5	4.17	P
CR2032-C45#	3.28	5	4.17	P



Supplementary information:

- No explosion and no fire.

6.5.6	TABLE: Test J: Free fall (Undischarged)		P
Battery No.	Before test	After test	Test results
	battery voltage (V)	battery voltage (V)	
CR2032-C46#	3.25	3.25	P
CR2032-C47#	3.26	3.25	P
CR2032-C48#	3.25	3.25	P
CR2032-C49#	3.28	3.28	P
CR2032-C50#	3.27	3.27	P

Supplementary information:

- No venting, no explosion and no fire.

6.5.7	TABLE: Test K: Thermal abuse (Undischarged)		P
Battery No.	Before test	After test	Test results
	battery voltage (V)	battery voltage (V)	
CR2032-C51#	3.28	3.28	P
CR2032-C52#	3.25	3.25	P
CR2032-C53#	3.23	3.23	P
CR2032-C54#	3.20	3.20	P
CR2032-C55#	3.26	3.26	P

Supplementary information:

- No explosion and no fire.

6.5.8	TABLE: Test L: Incorrect installation (Undischarged)	P
Battery No.	OCV of reversed cell, (Vdc)	Test results
CR2032-C56#	3.25	P
CR2032-C57#	3.23	P
CR2032-C58#	3.25	P
CR2032-C59#	3.26	P
CR2032-B60#	3.24	P

Supplementary information:

- No explosion and no fire.

6.5.9	TABLE: Test M: Overdischarge (Predischarged 50%)			N/A
Battery No.	OCV prior to discharging, (Vdc)	Maximum discharge current, (A)	Time for discharging, (hours)	Results

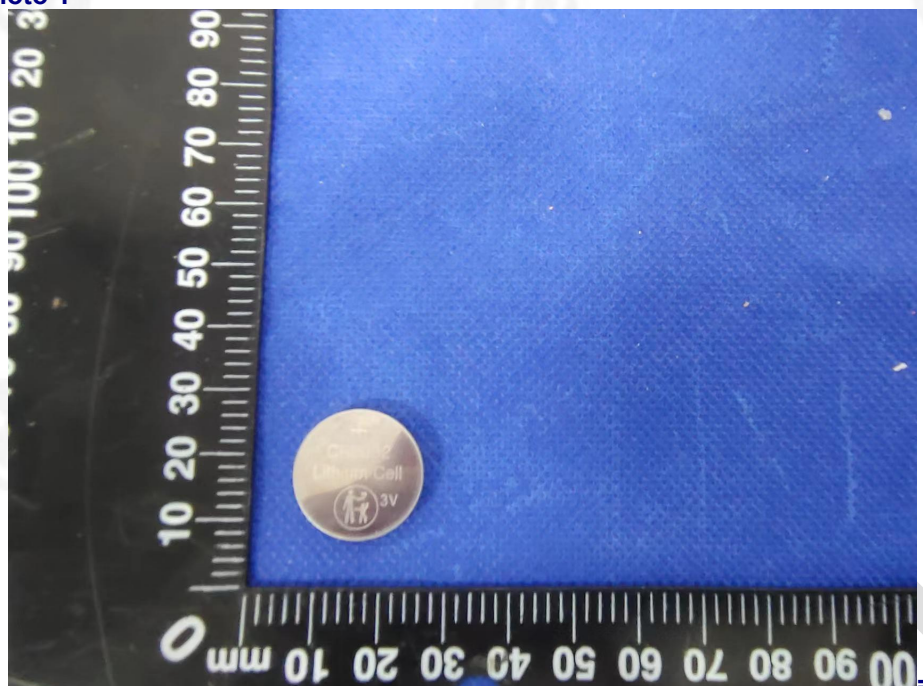


6.5.9	TABLE: Test M: Overdischarge (Predischarged 75%)			
Battery No.	OCV prior to discharging, (Vdc)	Maximum discharge current, (A)	Time for discharging, (hours)	Results
Supplementary information: - No explosion and no fire.				

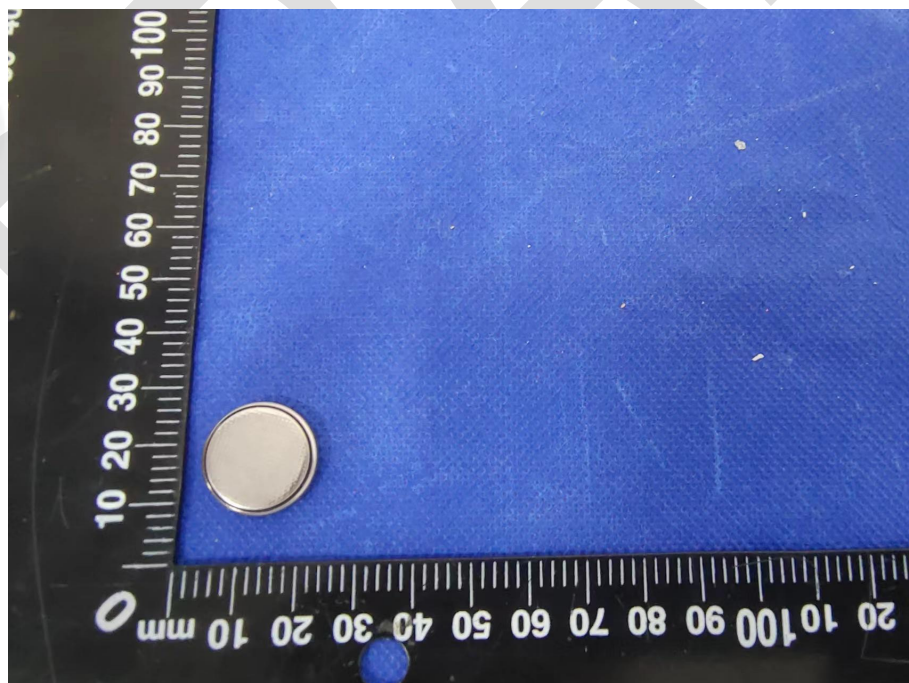


Attachment I: Photo-documentation

EUT Photo 1



EUT Photo 2



***** END OF REPORT *****